Feedback loops and climate change

In recent years, accelerating and decelerating warming trends have brought the idea of feedback loops to the fore in climate change science. In 1988, the climate scientist James Hansen spoke at a US congressional hearing of the rises in global temperature caused by human activity. He has since voiced the belief that the continued burning of fossil fuels could set in motion a series of calamitous positive feedbacks on Earth's climate, leading to the "runaway greenhouse" he describes in his 2009 book *Storms of My Grandchildren*.

One warming feedback loop is created by the melting of polar ice caps, as newly exposed land and water absorb the heat that the ice once reflected back into the atmosphere. The melting of Siberian permafrost is another warming loop. As temperature rises melt the permafrost, huge amounts of methane, a greenhouse gas, could be released into the atmosphere, accelerating global warming.